

Capstone Design

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Human AI Workflow

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Bias and Representation

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Capstone requirements and corpus. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone requirements and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

6. Solve the given case

Capstone: answer from one approved chapter with citations and a missing-evidence fallback.

Define acceptance tests.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Freeze the Study Buddy v1 design and acceptance tests before implementation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A complete capstone plan is not evidence that the implementation meets it; evaluate after building. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-evidence test.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Capstone Design

Plan a bounded study buddy around a small approved vocabulary corpus.

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Capstone Design

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define contract

Answer only glossary-definition requests supported by V1.



2. Plan normal test

"What is a stem?" must point to V1's stem entry.



3. Plan missing test

"How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.

CHANGE ONE THING

The corpus gains a teacher-approved flower definition in V2.

Explain the changed result using the matching lesson.

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan a bounded study buddy around a small approved vocabulary corpus.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

1. Plan missing test
2. Plan normal test
3. Define contract

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Plan missing test “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The corpus gains a teacher-approved flower definition in V2. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Add a flower test using V2 and record the corpus version.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Add a flower test using V2 and record the corpus version. The evidence boundary changed explicitly; the earlier corpus could not support this new entry.

Repair a missing building block

Review Human AI Workflow (ai-3/32). Goal: Explain the mechanism and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the mechanism and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the mechanism and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

